

Matrix Theory: Lecture 8

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Outline

- Norms in finite dimensional vector spaces
- Equivalence of norms
- Matrix and Operator norms
- The Condition Number



Norms

Def. Let V be a complex or real vector space. A **norm** $\| \cdot \|$ in V is a function $V \rightarrow \mathbb{R}$ which has following three properties:

- ① $\|v\| \geq 0$, and $\|v\| = 0 \Leftrightarrow v = 0$,
- ② $\|\alpha v\| = |\alpha| \|v\|$, for any $\alpha \in \mathbb{K}$,
- ③ $\|v + u\| \leq \|v\| + \|u\|$ for any $u, v \in V$ (**triangle inequality**).



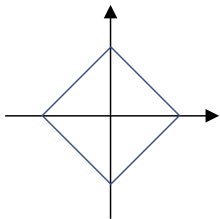
Norms

Ex. If $p \geq 1$, and $v = (x_1, x_2, \dots, x_n)$ then

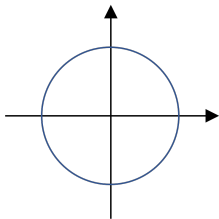
$$\|v\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

Norm:

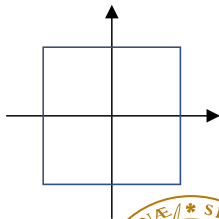
$$\|v\|_1 = \sum_i |x_i|$$



$$\|v\|_2 = \sqrt{\sum_i |x_i|^2}$$



$$\|v\|_\infty = \max_i |x_i|$$



Unit ball:

Remark: Depends on the choice of basis.



Theorem 11.7

For any norm we have:

- ① $\| -v \| = \| v \|,$
- ② $\| \sum_{i=1}^k \alpha_i v_i \| \leq \sum_{i=1}^k |\alpha_i| \cdot \| v_i \|,$
- ③ $\| u - v \| \geq \| u \| - \| v \|$ and $\| u - v \| \geq \| v \| - \| u \|.$

Or equivalently $\| u - v \| \geq \left| \| u \| - \| v \| \right|$ (Reversed triangle ineq.)



Convergence in Norm

Def. We say that the sequence of vectors v_k **converges** to the vector v **in the norm** $\|\cdot\|$ if the sequence of numbers $\|v_k - v\|$ tend to zero.

Theorem 11.8

- Every norm is a continuous function.
- For every norm $\|\cdot\|$ there exists positive constants c, C such that

$$c\|v\|_1 \leq \|v\| \leq C\|v\|_1$$

for any vector v .

- All norms are equivalent in the sense that any sequence of vectors v_k that converges to the vector v in one norm converges to the same vector in any other norm.

Holds in finite dimensional spaces.



Matrix Norms

Matrices of a fixed size form a vector space. We can use vector norms on matrices.

Ex.

$$\|A\|_{(p)} = \left(\sum_{i,j} |a_{i,j}|^p \right)^{1/p}.$$

(We write (p) to distinguish from operator norms, soon to be defined...)

The Frobenious norm is $\|A\|_F := \|A\|_{(2)}$.

Theorem 11.9

The following is true for the Frobenius norm:

- $\|X\|_F = \|X\|_{(2)} = \|X\|_2 = |X|$ for any column (or row) vector X .
- $\|AB\|_F \leq \|A\|_F \|B\|_F$ (the sizes of matrices A and B can be different).
- $|AX| \leq \|A\|_F |X|$ for any column vector X .

Operator Norms

Let V be a vector space with a norm $\|v\|$ and $\mathcal{A} : V \rightarrow V$ be some linear operator. We define the **operator norm** of $\mathcal{A}$ as

$$\|\mathcal{A}\| = \max_{v \neq 0} \frac{\|\mathcal{A}v\|}{\|v\|}.$$

- Does to maximum exist?
- Is this a norm?
- How do we compute it?

Theorem 11.10

The operator norm $\|\mathcal{A}\|$ has the following properties:

- 1 $\|\mathcal{A}\| = \max_{\|u\|=1} \|\mathcal{A}u\|.$
- 2 $\|\mathcal{A}v\| \leq \|\mathcal{A}\| \cdot \|v\|$ for any $v \in V$ (equality is achieved at least for one non-zero vector).
- 3 $\|\mathcal{A} + \mathcal{B}\| \leq \|\mathcal{A}\| + \|\mathcal{B}\|$ for any operators $\mathcal{A}, \mathcal{B}.$
- 4 $\|\mathcal{A}\mathcal{B}\| \leq \|\mathcal{A}\| \cdot \|\mathcal{B}\|$ for any operators $\mathcal{A}, \mathcal{B}.$

The p-Operator Norm

With $\|A\|_p$ we mean the operator norm obtained from the p-vector norm, that is,

$$\|A\|_p = \max_{X \neq 0} \frac{\|AX\|_p}{\|X\|_p}.$$

Theorem 11.11

Let $A = (a_{ij})$ be an arbitrary complex $m \times n$ matrix and $A^H = \overline{A^T} = (\overline{a_{ji}})$ its **Hermitian conjugate**. Then

- $\|A\|_1 = \max_j \sum_i |a_{ij}|$ (the maximal 1-norm of the columns)
- $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (the maximal 1-norm of the rows)
- $\|A\|_2 = \sqrt{\lambda}$, where λ is the largest eigenvalue of $A^H A$.
- $\|A\|_F = \|A\|_{(2)} = \sqrt{\text{tr } A^H A}$ and $\|A\|_F \geq \|A\|_2$.



The Condition Number

Let A be an invertible square matrix.

Def. The **condition number** $\kappa(A)$ is defined as

$$\kappa(A) = \|A\|_2 \cdot \|A^{-1}\|_2.$$

Note that

$$\kappa(A) \geq \|AA^{-1}\|_2 = 1.$$

Matrices with small condition numbers are preferable in numerical calculations.

If $Ax = b$ and b is affected by a perturbation δb , what is the size of the perturbation δx ?

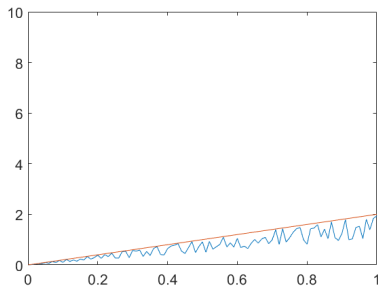
$$\frac{\|\delta x\|_2}{\|x\|_2} = \kappa(A) \frac{\|\delta b\|_2}{\|b\|_2}.$$



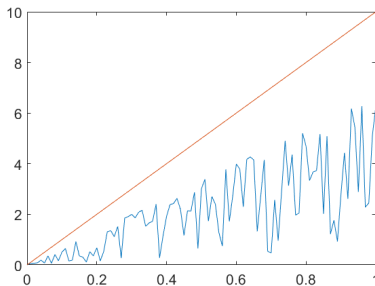
The Condition Number

Ex.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \kappa(A) = 2$$



$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix}, \kappa(A) = 10$$



$\frac{\|\delta b\|_2}{\|b\|_2}$ (x-axis) vs. $\frac{\|\delta x\|_2}{\|x\|_2}$ (y-axis)

red: worst case bound, using $\kappa(A)$

blue: random b and δb .

